

BRATT'S LAKE, SK, Canada

WMO: 715690

Lat: 50.201N

Lon: 104.710W

Elev: 1903

StdP: 13.71

Time Zone: -6.00 (W06)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-25.9	-21.1	-32.5	0.9	-25.9	-27.3	1.3	-21.0	32.3	12.1	29.2	12.6	9.9	270	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.3	87.3	63.3	83.6	62.3	80.4	61.2	68.2	79.0	66.0	77.3	64.2	75.5	11.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	97.8	73.1	62.1	89.6	70.9	60.0	83.1	68.9	33.8	79.1	31.9	77.1	30.4	76.1	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.6	26.0	23.3	DB	-33.5	94.8	5.5	3.6	-37.4	97.4	-40.6	99.5	-43.7	101.5	-47.7	104.1	(4)
(5)			WB	-33.9	72.9	5.2	2.7	-37.7	74.9	-40.8	76.5	-43.7	78.0	-47.5	80.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	8.5	7.0	21.3	37.3	51.3	60.7	65.1	63.3	54.4	39.8	24.4	9.5		
	DBStd	23.87	14.86	15.03	14.64	10.34	8.64	5.86	5.21	5.89	8.22	8.59	12.23	14.30		
	HDD50	6265	1286	1203	889	391	91	4	0	0	45	332	768	1256		
	HDD65	10358	1751	1623	1354	831	429	152	64	104	331	780	1218	1721		
	CDD50	1537	0	0	0	10	132	324	467	412	175	17	0	0		
	CDD65	155	0	0	0	0	5	21	67	51	11	0	0	0		
	CDH74	2721	0	0	0	10	226	397	861	882	339	6	0	0		
	CDH80	786	0	0	0	1	56	89	246	285	109	0	0	0		
(14) Wind	WSAvg	11.6	13.1	11.6	12.6	12.8	12.7	11.3	9.6	9.3	10.8	11.9	12.1	11.7		
(15) Precipitation	PrecAvg	15.1	0.5	0.4	0.6	1.0	2.0	3.0	2.3	2.1	1.1	1.0	0.5	0.5		
	PrecMax	21.8	1.5	0.7	1.9	2.8	5.1	6.5	5.8	4.5	2.6	3.1	1.4	1.5		
	PrecMin	9.2	0.0	0.0	0.0	0.2	0.3	0.9	0.7	0.1	0.0	0.0	0.0	0.0		
	PrecStd	3.2	0.3	0.2	0.4	0.7	1.2	1.4	1.1	1.1	0.7	0.7	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	41.7	59.1	73.7	86.5	88.3	91.2	92.6	88.7	74.0	61.0	42.4		
		MCWB	35.3	36.1	46.7	51.3	57.1	62.1	68.2	63.5	61.2	54.4	46.2	34.9		
	2%	DB	36.5	36.4	51.5	66.4	80.2	82.0	86.3	86.7	83.1	65.9	50.0	37.0		
		MCWB	33.3	33.0	43.2	47.5	56.1	61.1	66.5	63.3	58.7	50.1	41.0	33.1		
	5%	DB	34.0	32.2	43.6	60.9	75.2	78.3	82.2	82.7	77.2	59.2	44.2	32.8		
		MCWB	31.7	30.4	38.5	45.1	53.2	59.8	64.7	62.8	57.0	47.5	38.1	30.5		
	10%	DB	29.9	28.5	37.9	55.1	70.4	75.2	78.7	79.0	71.5	54.4	39.6	28.5		
		MCWB	28.0	26.8	34.6	42.3	51.4	58.0	64.0	61.5	55.2	45.0	35.5	26.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.6	36.5	47.3	52.6	60.9	68.8	73.4	70.0	64.0	56.3	47.8	35.8		
		MCDB	39.3	40.5	56.6	70.6	75.7	79.7	84.3	79.7	81.7	69.0	61.2	40.9		
	2%	WB	33.6	33.3	43.0	48.9	57.6	64.7	69.3	67.0	60.8	51.9	42.1	33.3		
		MCDB	36.1	36.1	50.8	63.6	73.0	75.1	79.9	78.9	77.4	62.1	49.0	36.8		
	5%	WB	31.5	30.8	38.4	46.1	55.4	62.5	67.1	65.0	58.2	48.8	38.5	30.4		
		MCDB	34.0	32.7	43.4	58.1	71.0	72.1	77.8	77.6	74.1	58.0	43.2	32.7		
	10%	WB	28.1	26.8	34.7	43.6	53.2	60.4	65.2	63.2	55.6	45.8	35.6	26.9		
		MCDB	29.9	28.6	37.7	53.7	66.4	70.9	75.9	75.0	69.0	53.3	39.3	28.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	19.1	18.1	22.6	26.9	23.0	24.3	26.6	26.2	21.6	18.3	16.5		
		MCDBR	19.8	19.1	22.8	35.1	35.8	29.5	29.7	34.0	37.1	30.9	25.5	19.1		
		MCWBR	17.1	15.6	15.0	19.0	16.4	13.3	13.7	14.6	16.0	17.4	17.0	15.8		
	5% WB	MCDBR	20.3	18.3	22.6	30.5	29.6	23.2	25.2	28.1	32.6	28.5	24.1	18.8		
		MCWBR	17.8	15.2	15.2	17.6	14.5	12.2	13.2	14.0	14.7	16.5	16.3	16.0		
(40) Clear-Sky Solar Irradiance	taub		0.240	0.309	0.272	0.328	0.347	0.351	0.359	0.369	0.327	0.280	0.250	0.237		
	taud		2.393	2.185	2.515	2.393	2.391	2.423	2.415	2.366	2.458	2.564	2.519	2.423		
	Ebn at Noon		264	259	302	291	289	287	283	274	276	276	261	249		
	Edh at Noon		19	31	27	34	36	35	35	35	28	21	17	16		
(44) All-Sky Solar Radiation	RadAvg		409	744	1166	1531	1846	1938	2095	1665	1238	745	412	293		
	RadStd		49	64	95	116	151	123	124	113	110	93	58	35		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page